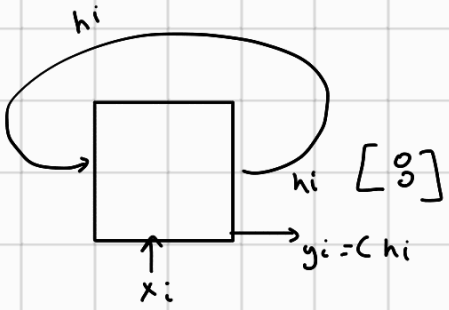
RNN





① $[B; A]\begin{bmatrix} x_i \\ h_i \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix}\begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} = h_i = h_1$

$C h_i = [\ ] = y_i$

② $\begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \end{bmatrix}\begin{bmatrix} 4 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 17 \end{bmatrix}$

$y_2 = 16$

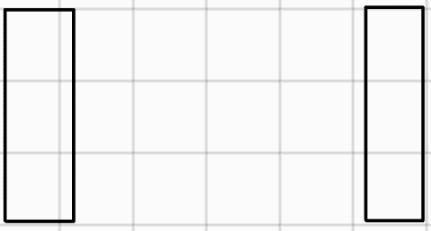
MSE = ?

$\hat{y} = f(C A \begin{bmatrix} x \\ h \end{bmatrix})$

$L = (\hat{y} - y)^2$

I run

I university bus run



Q = Query

k = key

V = Value

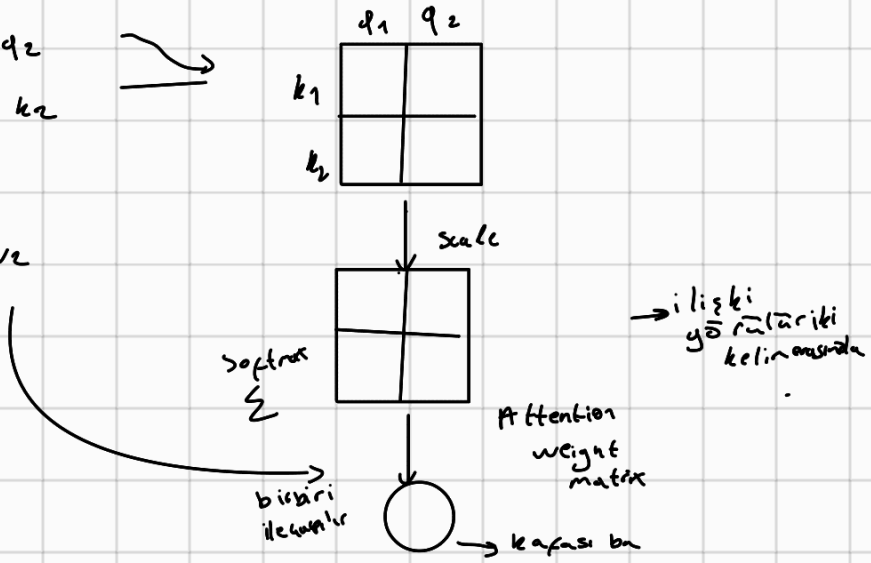
attention = $\text{softmax}\left(\frac{k^T \cdot Q}{\sqrt{d_k}}\right) V$

↓ kare kökü

$W_Q \cdot w = q_1\ q_2$

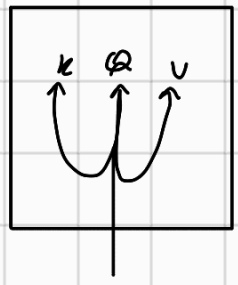
$W_K \cdot w = k_1\ k_2$

$W_V \cdot w = v_1\ v_2$



Bunun adı self attention

Makalelerde böyle görürsünüz



→ multi headde ole olabilir

Q: Ne aradığımız

k: Ne ile ilgili

V: Neyi kullanacağız

semt ile tuğçim adını karıştırdık



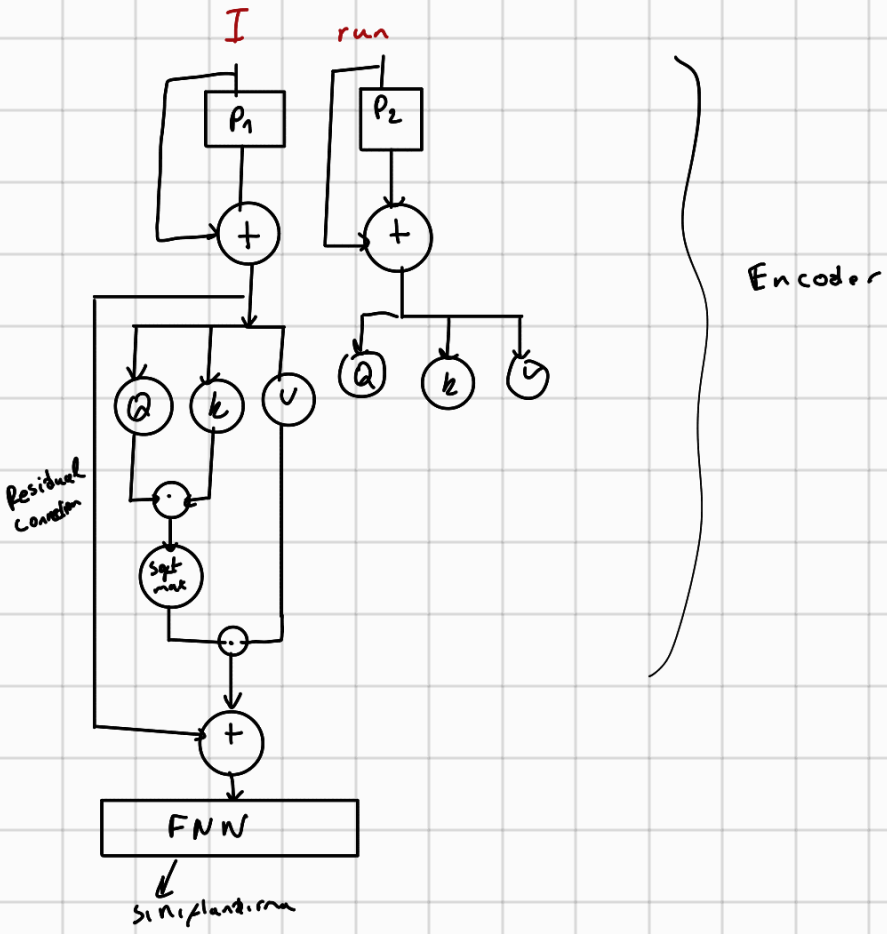
bunun $\times W_1$ → positional encoding matris

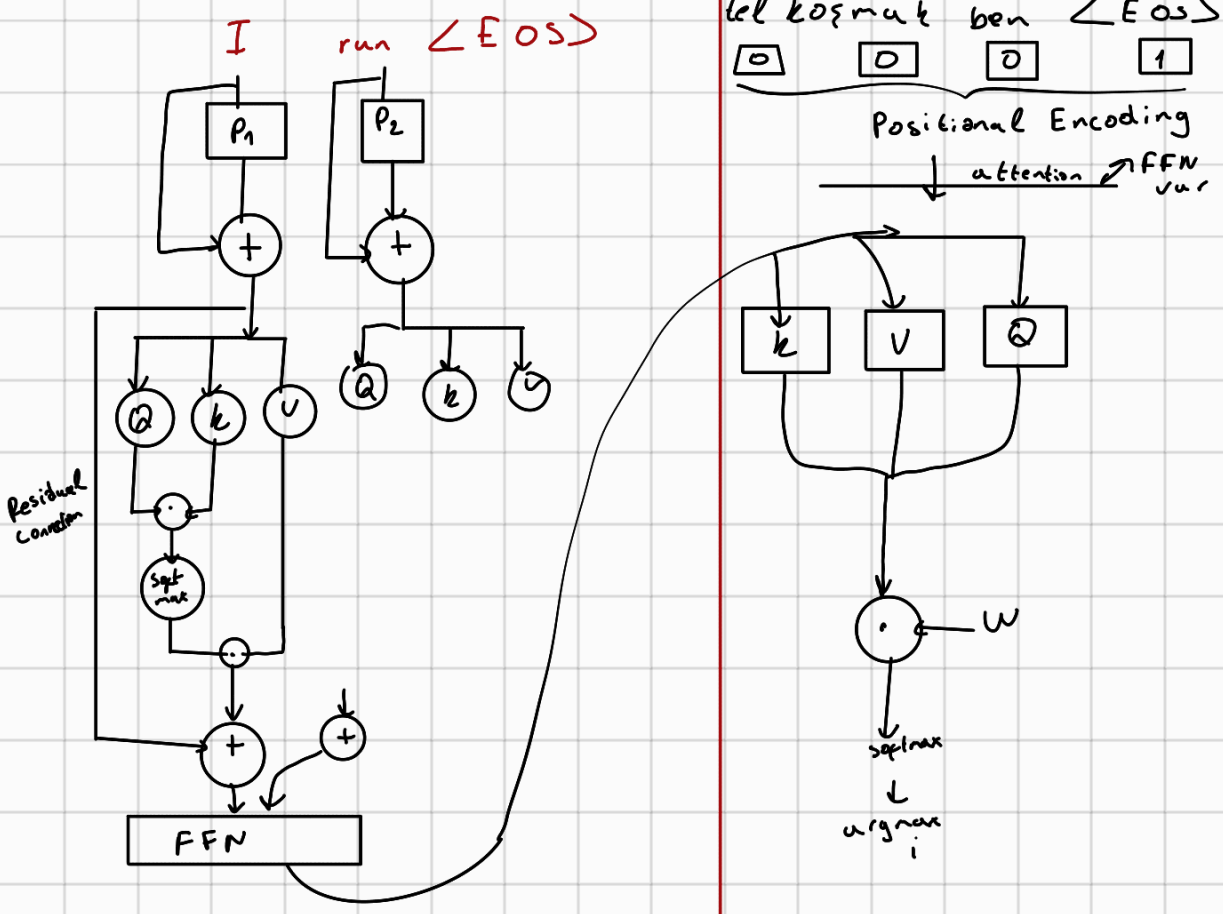
$x_I + W_1 x_I = \bar{x_I}$

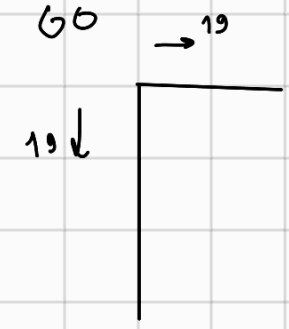
positional embedding

I run

I university bus run







361! ihtimal

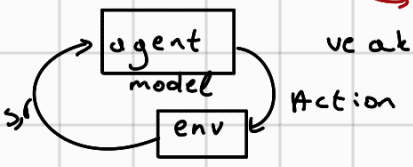
2^{361} $\pi(a|s)$

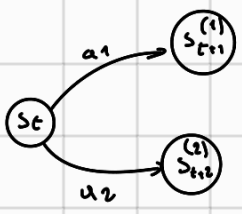
$\pi(s) = a$

$\pi \neq \Rightarrow V_\pi$ → optimum policy

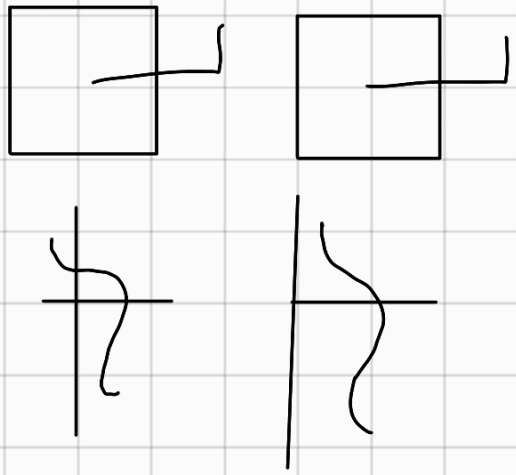
$P(s_{t+1}, a_{t+1} | s_t, a_t)$

ve aksiyon eylem söyleneli:





- Model tabanlı
- Modelsiz



Exportion
Exploitation
Pilemenc

$$V(s) = E\{G_t | S_t\}$$

$$G_t = \sum_{n=1}^{T} r_{t+n}$$

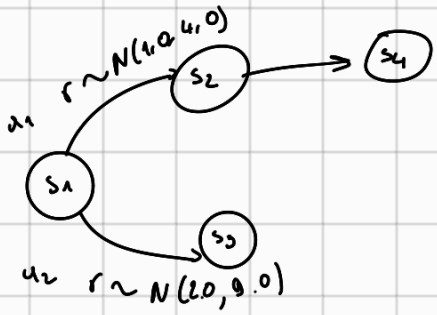
$V(s) = E\{G_t | S_t\}$ nasıl hesaplanır

$s_0, a_0, r_1, s_1, a_1, r_2 \ldots$

Gelecekte alacağım ödüllerin ortalaması

$$V(s) = E\{G_t | S_t\}$$

$$Q(s,a) = E\{G_t | S_t, a\}$$



(1)

$Q(s_1, a_1) = 0$

$Q(s_1, a_2) = 0$

SARSA

$0 < \alpha < 1$

$$Q(s_1, a_1) = Q(s_1, a_1) + \alpha\left(r + V(s_2) - Q(s_1, a_1)\right)$$

$Q(s_2)(s_2)$

r: reward

$= 2$

Q learning

Policy gradient theorem

Actor critic (value, policy birleşimi)

Bunun dışında

Multi agent re inforcment learning

$$G_t = \sum_{n=1}^{t} \gamma^{n-1} r_{t+n} = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3}$$

$$0 < \gamma \leq 1$$

$$= r_{t+1} + G_{t+1}$$

$$V(s,t) = r_{t+1} + V(s_{t+1})$$